

Making an ArcGIS Story Map

The focus of this manual is the ArcGIS StoryMaps, an online application offered by ArcGIS. ArcGIS StoryMaps is a place-based digital story-telling tool. In history it can be used to create narratives, explore places as well as a public outreach for your research. Before jumping on the application though you should structure your story, figure out who is the audience and what is the message you want to convey. For tips on storybuilding you can follow the [link](#).

ArcGIS access

The University of Groningen has access to the software (both the online tools and the desktop applications such as ArcGIS). For access to the online tool-suite of ArcGIS open the home page of arcgis. Follow the [link](#) or type: **www.arcgis.com**.

Sign in and type into the ArcGIS organisation:
 RUG (like in Figure 1) and click continue.
 Afterwards go to the RUG button and log in with
 your RUG account.

Now you have an account with access to the
 different tools the ArcGIS provides. In the future
 when you want to log in you can type in the search
 bar **rug.maps.arcgis.com** which will skip the steps
 in the general home page of arcgis and will open
 directly the RUG Geoportal. To access your
 account click on ‘Sign In’ and sign in the geoportal
 by selecting RUG and then log in your student
 account.

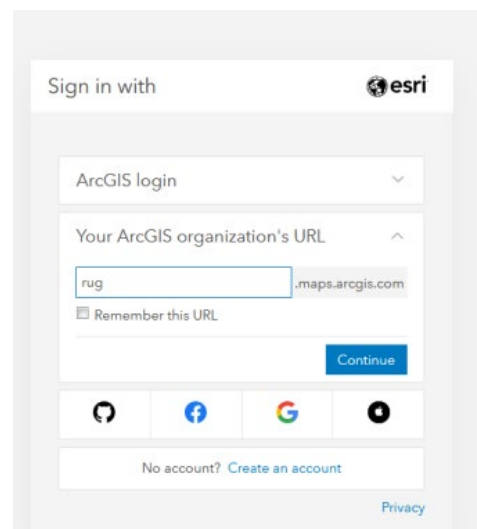


Figure 1 Sign in process with RUG

To explore the tools the ArcGIS offers click on the icon with the dots next to your account
 name and locate the StoryMaps icon.

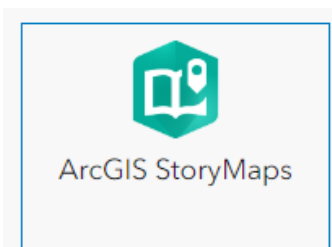
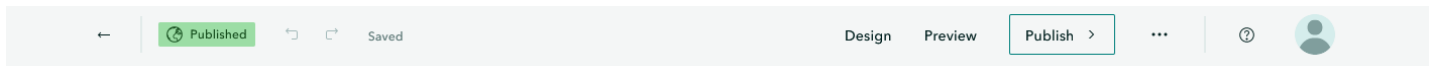


Figure 2 StoryMap

Building in StoryMaps

Most likely your initial page is empty, to start making your StoryMap go to the top of the page click on the ‘+New Story’ and select **start from scratch**. This is the best option to get accustomed to the tools and be able to decide which one suits you better. At the end of this tutorial there will also be instructions for the templates.



Interface

Figure 3 Interface

Before you do anything you should familiarize yourself with the interface (Figure 3). On the very top, starting from left to right you see:

- the status of the StoryMap (draft, published, unpublished changes)
- an undo and redo button, it does not work with the standard abbreviations for those actions.
- A text that says either ‘Saved’ or ‘Saving...’ the program automatically saves your process.
- ‘Design’: This tab includes options on what the story will look like and how the viewer will navigate through it. More options for the design will be given later on, you should explore it when you already have some content on.
- ‘Preview’: The tool allows the creator to enter a reader mode. Use it to see how the design and the pieces of the content will look for the reader.
- ‘Publish’: As it is already stated, the story is only visible to you, and it is set on private. When you are done with your story then you should change this setting to ‘Public’. It is possible to change the status of your story at any point.
- A dotted icon includes specific settings about the information shared in the published story.
- A question mark icon leads to the documentations and the help provided by the creators of the software.

Content

Now you can start creating your StoryMap, first give a title to your story and add a cover image. You can choose how much of the cover you can see by clicking on the ‘Design’ tab. Moreover, it is possible to change the focal point of the figure as well as the extent of the figure visible.

Under the title you can add content. By clicking on the highlighted + sign the program offers a variety of tools to create content. The tools are divided in three groups (basic, media, immersive) (See Figure 4).

You can explore on your own the options by creating new pieces and delete them afterwards. You can also preview how the story will look for the reader. In the next pages, there will be a detailed description of the use of tools and some examples of their use in history. If you want to look at more details on the program and the tool you can follow this [link](#) that includes a presentation of the tools.

Since you are building the StoryMap from scratch it is essential to have a good planning on what you will present and how. The program offers so many alternatives that it is very easy to

become chaotic. For this reason, the first step of using the StoryMaps is to create a clear storyboard.

Basic Tools

The tools here are the starting tools for your story are the basic ways of adding text information and hyperlinks. **Text** allows the creator to write free text or copy paste parts of text. The program allows you to formulate it like a word file offering formats by double clicking on your text (see Figure 4 bottom part). By using headings, and an activated navigation mode (see in Design), the headings are turned into hyperlinks in the text. The option '**Button**' creates and embedded url link for an external website and write a text to provide information for the link. Be aware that there is a word limit for the text. The **separator** creates a line to separate visually the parts of the text. In the latest version, you can also add **code** and **table**.

Media Tools

This set of tools can turn your Story Maps from a plain text into a multimedia experience. You can create multiple layers to explain, present, show and add dimension to your text. Of all the options the ones that are presented here are: map, image, embed and swipe.

- The **image** media allows the user to add figures that can be used to explain, enhance and illustrate the text. A pop-up window opens where you choose the image you want to add to your StoryMap. You can either browse your computer or drag and drop. This tool only allows you to display one image. The user can change the size and the location of the figures (see instructions below), however, the images are not interactive.

Edit Images.

Now StoryMaps allows for some limited image editing within the StoryMap environment. This is still in BETA phase, so there is a possibility that it will crash or too slow, just refresh the page or go back to the Stories page. The editing options are limited and the “crop” and “markup” tool.

There are three ways to use the **crop tool**

1. adjust the corners of your image as a rectangle, rounded rectangle and circle.
2. crop the image with different methods.

Original crops diagonally.

Freeform crops independently from the points around the image.

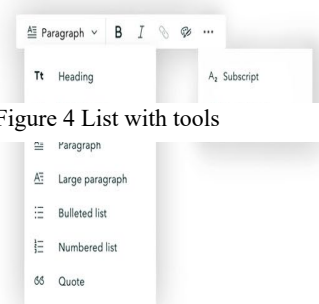
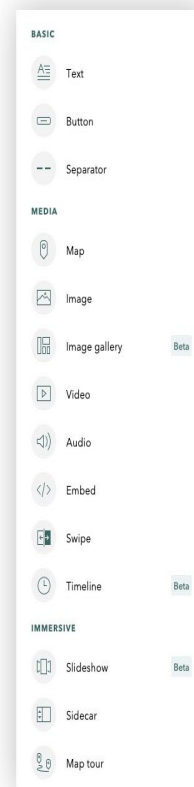


Figure 4 List with tools

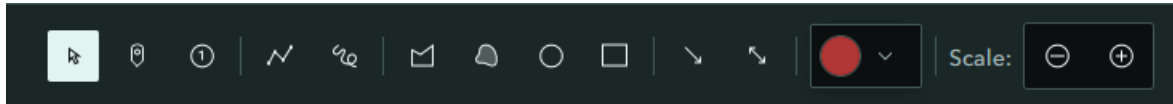
Square it selects a square in the image and crops accordingly.

3*2, 4*3 and 16*9 crops the image accordingly.

3. Rotate the image per 90 degree clockwise.

4. Flip horizontally, which mirrors the image on the horizontal axis.

The **Markup tool** allows the users to add shapes on the image, including points, lines, areas and shapes as well as arrows. It is possible to change the colour and size of the object by “scale”. To edit use the “select” tool



To go back to the start select the three dots icon on the top right corner of the page and select “Revert to Original”. “Cancel” cancels all the changes that were made, both with the crop and markup tool. “Save” saves the changes and returns to the StoryMap.

Example

An example on how to use the editing tools is the following picture which was converted into this: [image example](#)



The image was cropped with the “Freeform” tool and cropped the top and the bottom of photo. The mark up tool was used to create a point to show the Kaikos river valley. The colour of the marker was changed to black to stand out in the image.

The edited image was added in the text as medium size image.

- The **map** tool allows you to add the location of the site you are presenting on the map. By clicking on the option you are transferred into another page where you are asked to add a map. There are different types of maps you can incorporate in your Story Map. Already created maps that are in your content or Express maps.
 - Content maps are maps with data you have already loaded into a map. The features we provide you with to load into content maps are network lines and the single inscription feature.
 - Go to ‘+ new express map’. When you open the tool you encounter an empty map of the whole world. To add points to the map you can either zoom in to your destination and click or use the search bar to locate the site you are working on.

After creating the point you can edit the feature, change the color of the pointer, add an image and a description. You can change the default settings of the map (basemap, legend, etc). To make those changes go to the option tab. The program offers a variety of basemaps that highlight different features. Have a look at the different option and consider:

- Which information should also be visible in the map?
- Do you want to include modern cities and roads?
- Is the landscape important for your research?

When pleased with the result, click ‘done’ for the edits on the point and then ‘done’ on the button on the map to save all the edits. Then the map with the point will appear on your StoryMap. If you had added an image will appear when you hover the cursor over the pointer and it is extend when you click on it. It is possible to return back to your map and make changes. Below you can see an example of the Map tool for showing the position of the mountain in Peloponnese.

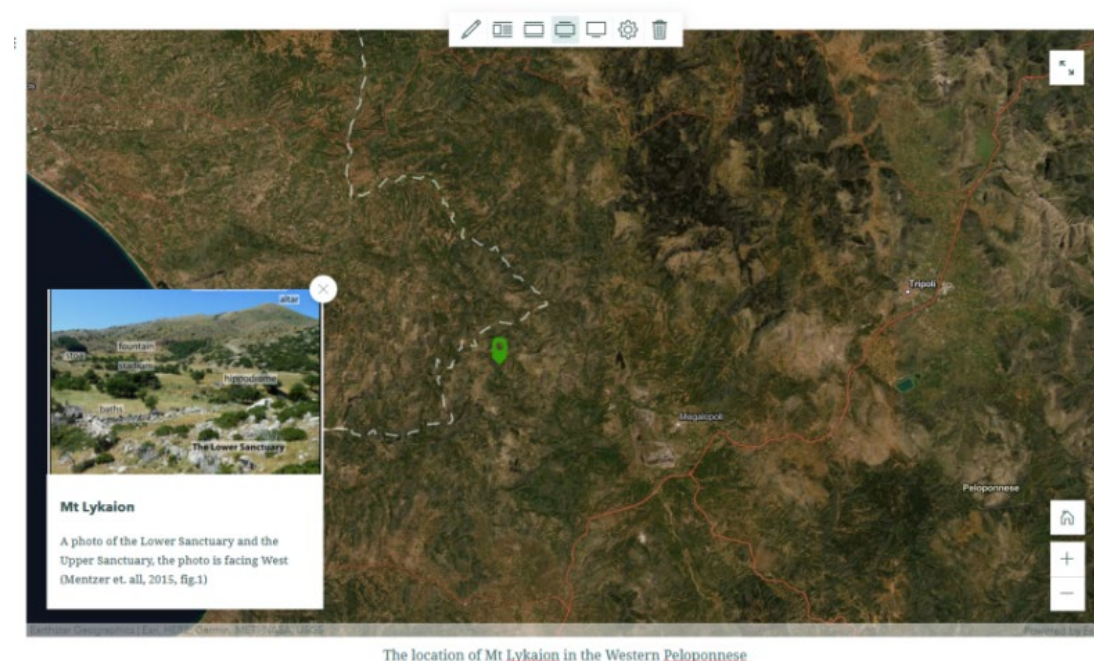


Figure 5 Example of the use of the Map tool with a figure and description on the figure, the basemap is set on Imagery (hybrid)

- The **embed** tool allows the user to add a url link with a automatically generated summary from the site. It is on the hand of the creator if they want the links to be accessible in the form of a button or in a form of a box. Below you can see the visual difference between the two options. The color of the button depends on the Design style of the StoryMap.

For more information

Mt. Lykaion Excavation and Survey Project

The Mt. Lykaion Excavation and Survey Project in Arcadia, Greece, is a long term, multi-disciplinary, scientific investigation of one of the most important Zeus sanctuaries in the ancient world.

<http://lykaionexcavation.org>

Figure 6 Difference between button and embed tool

- Another media tool is the **swipe** tool. This tool allows the creator to add two pictures or web maps. The images are overlapping and the swipe tool allows the reader to see more from one picture and less from another. A way of using this tool can be for an apple to apple comparison of the architecture of two temples, in artifacts or even between maps. If you use the swipe tool the figures might be automatically cropped. To avoid that, go to the options and click on the display tab 'fill(do not crop)'. To see how the swipe tool looks in the text go to the end of this manual (Navigate a StoryMap) and see how it looks in context.

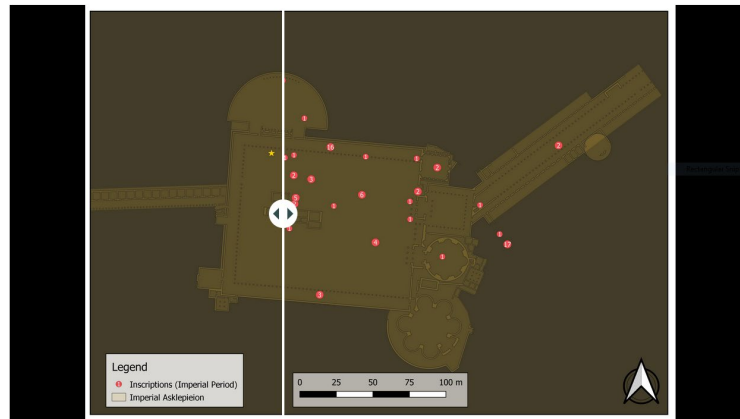


Figure 7 Example of a swipe tool with fill (do not crop) with black background with black color. The left image is an individual feature, the image on the right are contemporary features.

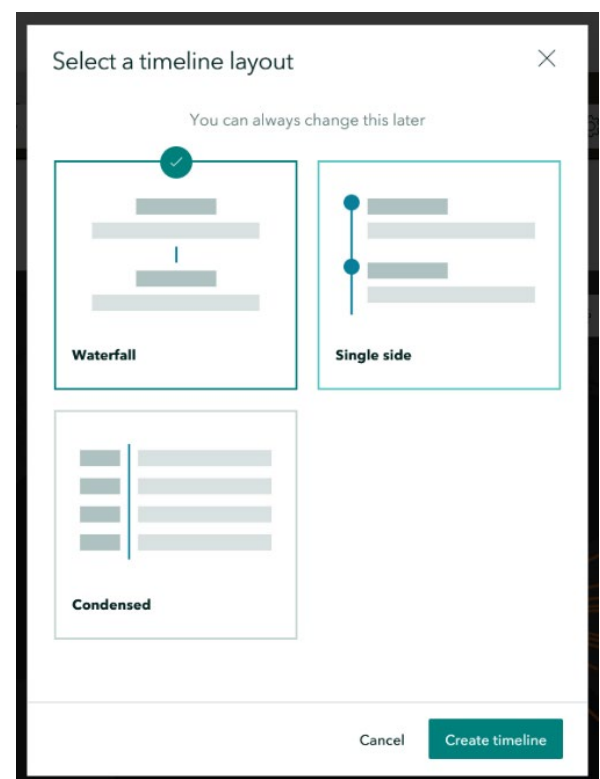
- The last media tool is the **timeline** tool. This tool allows the creator to present the timeline of events in an interactive way. There are multiple layouts to for a timeline. When you select this feature you write the temporal part (time or date) as well as the text part, the description. The combination of the temporal and text part are called “events”. When you want to add more events on your timeline you select the “+” sign and you are given the option between “event” and “spacer”. Spacer is an intermediate feature between events.

Apart from text it is also possible to include media in the events. Move the censor all the way to the right and select the “add image”.

Layout and size

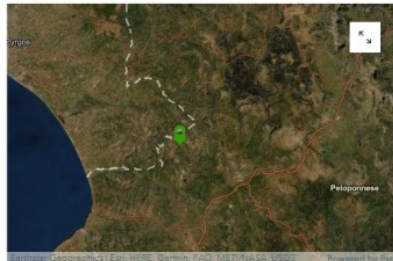
The media and the basic tools can be combined. Basic creates static features and the media features can be intertwined with them. By altering the position and the size of the media feature it is easier to read through the text. It is possible to change the location of the feature and move it up and down in relation to the other features. Move the censor to the top-left side of the feature, over the icon with the four dots and click and drag the feature.

Moreover, you can change the size of the images. When you move the censor on the media features, some icons appear on the top of the figure. There are the options small, medium, large which correspond to how much space the figure covers in the screen. The float option is smaller in size but allows the creator to include it in the text. Below is an example of a combination of text and image.



Excavation History

Mt Lykaion due to its significance in the Ancient World, it has been extensively excavated. The site was first excavation campaigns in the site took place in the late 19th and early 20th century and provided the first evidence of dedications from the mountain-top shrine. The ash altar was the area used to burn the dedications of animals and several ancient authors including Plato, Theophrastus and Pausanias mention human sacrifice at the site.



The location of Mt Lykaion in the Western Peloponnese

the votives from the ash altar.

For more information

The mount has been revisited numerous times by different excavation campaigns that attempt to answer questions about the origins of Greek cult and Greek athletics and is sponsored by multiple academic institutes such as the University of Arizona and the Greek Archaeological Service under the auspices of the American School of Classical Studies at Athens and the Ministry of Culture and Sports of Greece.

The site includes two main parts the upper and the lower sanctuary. Impressive highlights include the hippodrome, the Sanctuary of Zeus, the temenos and

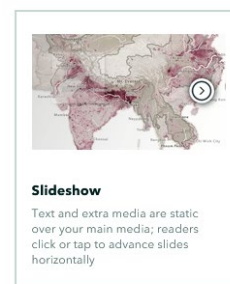
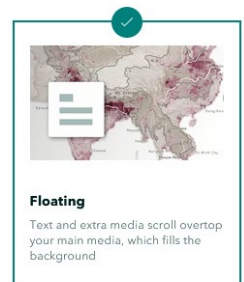
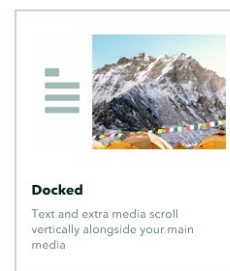
Figure 9 Example of a float feature (map) to describe the location of the feature that the text describes

Immersive Tools

The tools in this group combine text and figures, they cover the entire screen and are multi-layered. Moreover, they are interactive for the audience. The reader is able to move around the content by clicking or scrolling. Below you will find a presentation of the pros and cons of using each tool as well as proposed uses for the stories. The most useful one for this assignment is the map tour (look further down).

Sidecar

This tool includes a visual and a textual component with three possible combinations: (1) floating when the panel is moving with the figures, (2) docked where the panel is a stable feature, and (3) Slideshow. The main difference between the formats is the navigation. In 1 and 2 the viewer can navigate the layout by scrolling, in 3, the reader clicks on the arrow to navigate horizontally.



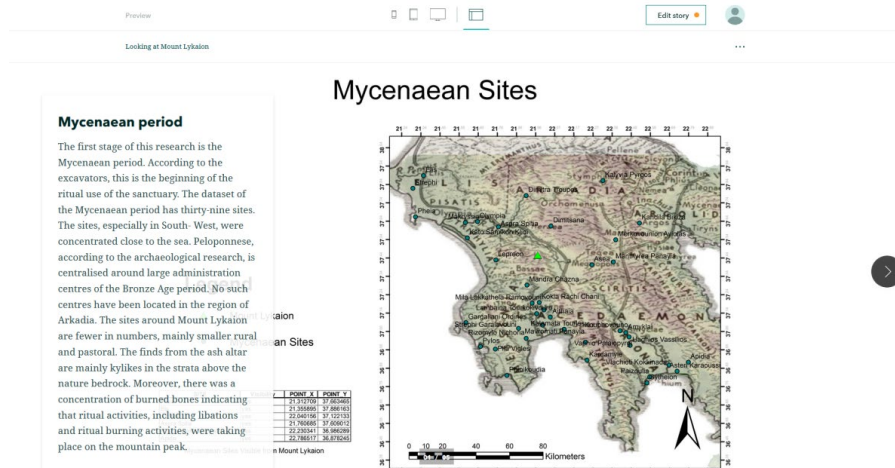


Figure 10 Slideshow in the preview mode with a figure with map and as a background. The theme is set on 'Summit'.

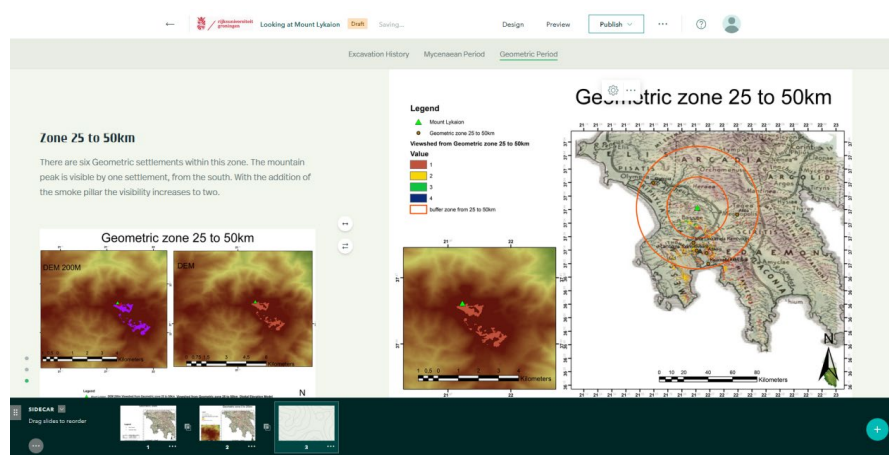


Figure 11 Sidecar with docked panel in edit mode. In the panel there is a subheading, text and a figure. In the media panel there is a figure map. The theme is set on 'Ridgeline'.

Map tour

Lastly, the final immersive tool is the **map tour**. This is the setting that allows the reader to navigate through a map and connect the location with text and images. This tool can be used the best to imitate the experience of giving a tour in real life. The creation of the location points is the same as in the media> maps part of the manual. The application offers a lot of options on how to create those tours (as always it is possible to change layout later on).

There are two main options and those have some sub-options:

- The first option is the *Guided tour*: the creator controls the path that the reader will follow. This option allows the reader to have a more structured narrative. The secondary options regard the media in focus - it is possible to have a larger map (map focused) or a larger media (media focused). The guided tour with media focused setting can be used to show in detail photographs of temples as well as their geographical location. It is proposed to use this to create the tour in the archaeological sites, as it is possible to control the route of the reader as well as the narrative that he follows. An example of guided tour is [The man, the inscription the network](#)
- The second option is the *Explorer tour*. In this case the creator provides the locations (either in a grid or a list, based on the secondary options), and the reader can decide on

their own which point they will explore. The main focus in this case is the map and the size of the figure is smaller. This setting is preferred if there are a lot of locations.

Based on the types of photos you have you can choose different ways to create a map tour. If the media are geotagged then the program can create a point automatically. If not then you should go for the scratch option. In this case you will create the point and then add the media on your own. You can zoom in as much as the map allows you. If you add multiple points in close geographic proximity the preview of the map will also be zoomed in, as opposed to the global map you see in the beginning.

Templates

As stated in the beginning of the manual, it is possible to use templates to create a StoryMap. There are three template options: Sidecar, guided or explore tour. The main difference is that when you open your new story there is already an immersive tool, depending on your choice, open. You can add before and after any of the other tools and media you prefer. We suggest you use this only when you are really set on using an immersive tool for your narrative, however, it will not save you a significant amount of time.

Navigate a StoryMap examples

You can have a look the [collection](#) of the StoryMaps we created in the context of the “Deep-mapping Asklepion of Pergamon” project.

You can also experience how the tools look for the user by navigating through a pre-made story. The [StoryMap](#) available is made to present the results of a research that was conducted by the author on the visibility from Mount Lykaion.

As opposed to your assignment it is more based on the presentation of figures (maps). The research includes several sections that are made with the different settings:

- Excavation History: Text and floating media (map), moreover there is a button feature as well as an embed url.
- Mycenaean Period: Text and Swipe
- Geometric Period: Sidecar with Docked panel
- Archaic Period: Sidecar with Floating panel
- Classical Period: Sidecar with Docked panel
- Hellenistic Period: Text, image, swipe
- Perception: Map Tour (guided tour, media focused and list)

